

Interpretation of slides

Slide 1

Slide 2

Slide 3: Outline

Introduction

Part I: CNN Classification & Ensemble Learning (Kidney US)

Part II: Annotation-Based Learning & YOLOv8 (Kidney & Lung US)

Challenges & Future Directions

Summary

Interactive Q&A breaks included.

Slide 4

Ultrasound offers several advantages: no radiation exposure, real-time imaging, and portability.

It provides immediate bedside information without exposing patients to ionizing radiation.

Because of its clinical value, ultrasound has been proposed as the fifth component of the physical examination, alongside inspection, palpation, percussion, and auscultation.

Today, handheld ultrasound devices can even be connected directly to smartphones, making point-of-care ultrasound more accessible than ever.

Slide 5

This photograph shows a prenatal ultrasound examination conducted in a Maasai community in Tanzania using an ultrasound probe made in Taiwan.

By combining a portable ultrasound probe with a laptop computer, ultrasound examinations can be performed even in the most remote regions of the world.

We hope that artificial intelligence will further expand access to quality healthcare and help reduce disparities in medical services worldwide.

Slide 6 A Growing Healthcare Challenge in Taiwan

- More than 90,000 patients on dialysis
- Annual medical expenditure exceeds NTD 53.3 billion
- Additional societal costs are not included

Kidney Health Begins in Childhood!

Slide 7 Kidney Diseases and Congenital Anomalies

- Chronic and progressive changes
- Often associated with subtle or nonspecific symptoms

Slide8: Importance of ultrasound

Not all kidney abnormalities can be detected by urine or blood tests.

Ultrasound remains an indispensable diagnostic tool.

Slide 9:

Portable ultrasound devices and internet access are now widely available. The question is: how can we efficiently detect abnormal findings from all these ultrasound images?

To solve this problem, we developed an AI screening module for kidney ultrasound.

The system can run either through a cloud-based network connection or locally on the computer where the ultrasound examination is performed.

After the images are acquired, the AI automatically analyzes them and immediately generates a screening report, helping the examiner identify potential abnormalities in real time.

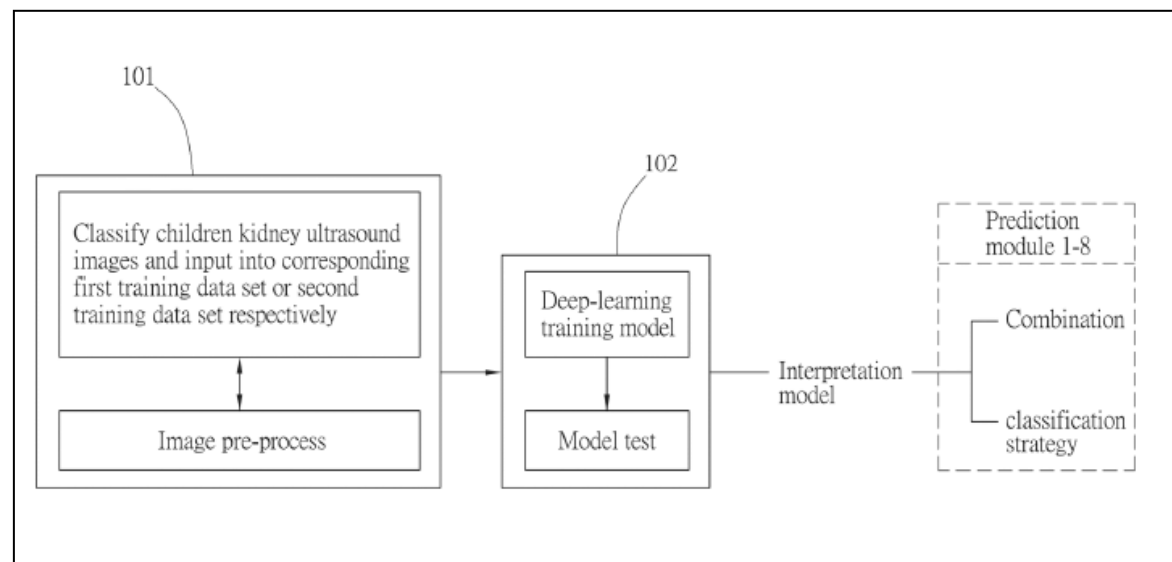
Slide10

Our AI-assisted interpretation system and workflow received patent protection in Taiwan in 2023 and in the United States in 2025.

Each patent includes seven claims that cover the complete workflow.

Slide 11: Short rest, feel free to participate the “Discussion”

Slide 12:



Slide 12

Based on clinical utility, I developed a kidney ultrasound dataset comprising five abnormal categories (including: hydronephrosis, cyst, stone, hyperechogenicity and space occupying lesion); and one normal category.

Slide 13 Ground truth establishment process

Image Collection Kidney ultrasound images were collected from PACS system and de-identified.

1. **Independent Review** Two pediatric nephrologists independently reviewed each image.
2. **Agreement Check** If both reviewers agreed, the image was included and labeled accordingly.
3. **Third-Expert Review** Discordant cases were reviewed by a third pediatric nephrologist.
4. **Final Classification** Clearly classified images were included in the dataset. Ambiguous images were separated for future analysis.
5. **Future Use of Ambiguous Images** These images will be used to investigate the impact of different labeling strategies on AI model performance.

Slide 15 Multiple Dataset Strategy

(continued) Step 5. Ambiguous Images

Some ultrasound images cannot be confidently classified as normal or abnormal. These ambiguous images are retained for further analysis rather than being assigned a forced label.

Step 6. Creation of Multiple Datasets

To evaluate the impact of different labeling strategies, three datasets are constructed. Dataset A uses a strict labeling strategy and includes only clearly defined normal and abnormal cases, excluding all ambiguous images. Dataset B uses a flexible labeling strategy in which ambiguous images are classified as normal. Dataset C uses an inclusive labeling strategy in which ambiguous images are classified as abnormal. Each dataset is then used independently for model training and testing.

Step 7. Comparative Performance Analysis

Model performance is compared across the three datasets. This approach allows us to evaluate the influence of ambiguous images and different labeling strategies on diagnostic accuracy and overall model performance.

Step 8. Selection of the Optimal Strategy

Based on the comparative results, the most appropriate labeling strategy is selected. The corresponding dataset is adopted as the final training dataset and serves as the foundation for subsequent model development and clinical application.

Key Message

By repeatedly testing models using datasets generated from different labeling strategies, we can better understand the impact of ambiguous images on model performance and establish the most appropriate ground-truth standard.

Slide 16

Original kidney ultrasound images undergo preprocessing and region of interest (ROI) extraction before entering the AI training workflow.

Slide 17 & 18 CNN=convolutional neural network

Before we move on to the AI section, I'd like to introduce my collaborator, Mr. Yueh-Chuan Chang and our workstation setting.

Slide 19:

It is an illustration showing each one of the first training datasets and its internal abnormal images in Step 101 of an embodiment of the present invention.

Slide 20 Deep Learning Model Architecture

- ResNet-50 backbone with ImageNet pretrained weights
- Feature extraction performed by the pretrained network
- Customized classification layers added for kidney ultrasound classification
- Fine-tuned using the annotated kidney ultrasound image dataset
- Visualized using Netron software

Slide 21:

Each prediction module independently evaluates the probability of abnormality. The final interpretation is generated through an ensemble strategy that combines multiple prediction modules.

Slide 22 Batch AI interpretation

- Automated analysis of 100 kidney ultrasound images
- Results exported automatically
- Processing time: approximately 30 seconds

Slide 23: Screenshot of the exported Excel results.

You can see the confidence scores generated by the AI for each module.

Each module makes its own prediction, and we combine these results to obtain the final classification.

Based on the combined scores, the image is classified as either normal or abnormal.

Slide 24: Multi-Module AI Interpretation Strategy

Each prediction module independently classifies an image as normal or abnormal. The final result is determined by combining the outputs of multiple modules through an ensemble strategy, improving overall diagnostic performance and reliability.

Slide 25: Our Study Published in *JMIR Medical Informatics*

- More than 20,000 kidney ultrasound images collected at TCVGH
- AI model entirely developed at TCVGH
- Deep learning system based on transfer learning
- Overall accuracy: 92.9% • AUC: 0.959

Slide 26: Number of normal images and five categories of abnormal images, together with the results of 5-fold cross-validation on the test dataset.

Slide 27: Validation Results

The model achieved favorable performance across multiple evaluation metrics, including sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and diagnostic accuracy, as summarized in the table.

Slide 28: External Validation Using Images from Other Hospitals

To assess model generalizability, kidney ultrasound images acquired from different ultrasound manufacturers at external hospitals were used for testing. Accuracy=19/20=95%

Slide 29: Clinical Applications in Real-World Practice

- Detection and surveillance of congenital kidney and urinary tract anomalies
- Evaluation of urinary tract infection (UTI)
- Investigation of acute abdominal pain
- Investigation of acute flank pain
- Monitoring of chronic kidney disease (CKD), including:
 - Increased renal echogenicity
 - New cyst formation

Slide 30 Dr. Hi from Bach Mai Hospital in Hanoi, Vietnam, previously visited our hospital for a three-month pediatric nephrology training program.

Later, I asked him to conduct a direct connectivity test using our AI-assisted interpretation system from the intensive care unit of his hospital.

Please watch the following video.

Slide 31

Slide 32: Next, I will show how AI models are trained to interpret kidney ultrasound images using the annotation and segmentation of anatomical structures and lesions.

Slide 33 To illustrate this method, I will begin with kidney ultrasound images as an example.

Slide 34 Annotation Workflow Using AnyLabeling

Slide 35 From annotation to model training

Slide 36 YOLOv8 Training Results

- Training and validation losses decreased progressively during training.
- Precision, recall, and mAP improved steadily with increasing epochs.

- The upward trend of mAP50 indicates improved detection and localization performance.
- Overall, the training process was stable and achieved satisfactory segmentation results.

Slide 37 AI Segmentation Demonstration

Please watch the following video.

This screen recording demonstrates the analysis of test kidney ultrasound images using the AI-based segmentation model developed by our team.

Slide 38:

The annotated results for each image are saved in the corresponding output folder.

Slide 39:

Please observe the comparison between the original ultrasound image and the abnormalities identified by the AI segmentation model.

Slide 40

From Classification to Anomaly Localization: Lung Ultrasound as an Example

Slide 41 The annotation of lung ultrasound images includes several abnormal patterns.

Slide 42: However, the interpretation of ultrasound artifacts is also an essential part of lung ultrasound examination.

Slide 43 B-line annotation using rectangular bounding boxes

Slide 44 Results of AI Detection of B-lines Using Rectangle Annotations

Slide 45: Lung Ultrasound Annotation : Pleural Line Segmentation,

Air bronchogram Detection

Slide 46: Air Bronchograms Are Better Suited for Segmentation Annotation

Slide 47: Comparison between segmentation and detection

Slide 48: short break for discussion

Slide 49: **Potential Impact of AI-Assisted Ultrasound Screening**

- Taiwan has 1,454 pediatric clinics serving approximately 3.8 million children
- Annual pediatric outpatient visits exceed 22 million
- Significant regional differences in healthcare resource availability

- AI-assisted ultrasound and telemedicine may facilitate earlier detection of abnormalities
- Improved access to medical care in rural and resource-limited areas

Slide 50: Remote AI-Assisted Ultrasound Analysis

- Hospitals in Vietnam can connect to our AI server in Taiwan via the Internet
- Ultrasound images can be analyzed remotely
- Anomaly screening results are available almost in real time
- Supports telemedicine and cross-border healthcare collaboration

Slide 51: Near-Future Applications of AI-Assisted Lung Ultrasound

- Beyond renal anomaly screening, lung ultrasound has strong near-future clinical applications
- AI-assisted ultrasound may facilitate earlier diagnosis of respiratory infections and pulmonary diseases
- Improved access to quality healthcare closer to patients' homes
- Reduced unnecessary travel and healthcare disparities
- Enhanced efficiency of healthcare delivery systems

Slide 52: Acknowledgements